

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	8.12/10
2021	CBSE(XII)	Pragati Public School, Delhi	91.4%
2019	CBSE(X)	St. Cecilia's Public School, Delhi	92.6%

Scholastic Achievements

- Secured **All India Rank 120** in **JEE Mains 2021** by scoring **295/300** with **100 percentile** in Mathematics and Physics
- Secured **All India Rank 2551** in **JEE Advanced 2021** among the 2 Lakh shortlisted candidates conducted by IIT Kharagpur
- Secured **All India Rank 181** in **KVPY 2020 SA** and **All India Rank 404** in **KVPY 2021 SX** conducted by IISc Bangalore
- Achieved **International Rank 2** in **International Reasoning and Aptitude Olympiad 2019** conducted by SilverZone
- Awarded **National Talent Search Scholarship 2019** by NCERT, secured **State Rank 3** in Stage 1 with a score of **191/200**
- Qualified for **RMO 2018, RMO 2019 & INChO 2020** conducted by Homi Bhabha Centre for Science Education, TIFR
- Among **National Top 1%** in **NSEJS 2017** and qualified for **Indian National Junior Science Olympiad(INJSO) 2018**
- Awarded **Junior Science Talent Search Scholarship 2018** by Delhi Govt, secured **State Rank 14** among 20,000 candidates

Key Projects

Algorithmic Graph Theory | Association for Computing Activities, IIT Kanpur (Mentor)  

(May'23 - Jul'23)

- Delved deep into **Graph Theory** with topics including **Graph Traversal, Minimum Spanning Trees, SSSP** and **APSP**
- Employed **Kruskal's & Prim's** algorithms for MSTs, **Kosaraju's & Tarjan's** algorithms for strongly connected components
- Solved complex problems in **Graph Coloring, Network Flow, Matching in Bipartite Graphs** and **Project Selection**
- Applied **Finite State Machines** and **Group Theory** to solve **Rubik's Cube, 3*3 Lights Out** and **Merlin's Machine**

Image Processing and Computer Vision | Stamatics, IIT Kanpur

(May'23 - Jul'23)

- Implemented image processing techniques for **Spacial Domain Transformation, Image Segmentation, Edge Detection**
- Built a **Jigsaw Solver** utilising **Canny Edge Detection** and designed **2D Dot Matrix Display** for random input numbers
- Applied image enhancement techniques to clear gutters of scanned images of books and enhance night images for theft detection
- Designed a **CNN** model for **multi-label image classification** leveraging **backpropagation** and **gradient descent** algorithms

The Flutter Effect | Association for Computing Activities, IIT Kanpur 

(May'23 - Jul'23)

- Developed a fully functional, robust, cross-platform **weather application** leveraging **APIs** for **real-time weather data sync**
- Utilised **FlutterSDK** using **Dart** for front-end development integrating **Flask** for backend and **MySQL** for data management
- Integrated **Firebase** for user authentication and implemented intuitive **UI/UX, location** services, and **caching** mechanisms

ML based Optical Character Recognition | Self Project  

(Apr'23 - May'23)

- Developed a sophisticated **Machine Learning** model for **Optical Character Recognition** using **LR, SVM, RF** classifiers
- Utilised hyperparameter optimization techniques like **k-fold cross validation, GridSearchCV** to achieve accuracy of **80.43%**
- Applied image processing techniques like **Erosion, Dilation, Thresholding, Histogram Equalization** and **Median Filter**
- Designed **CNN Sequential Architecture** consisting of multiple convolutional layers, pooling layers, and fully connected layers

Cervical Cancer Screening Prediction using Python - ML Modelling | Self Project 

(Feb'23 - Mar'23)

- Built **ML** models to perform **binary classification** for 4 target variables on the **Cervical Cancer Risk Factors Dataset**
- Preprocessed data by **Imputation, Normalization, Encoding** and visualization by **Boxplot, Heatmap, Confusion Matrix**
- Performed feature extraction using **PCA** and data balancing using **SMOTE** to improve accuracy to **94%** in **KNN** and **SVM**

AWS DeepRacer Student League | Team ViSiON, IIT Kanpur

(Nov'22 - Dec'22)

- Developed **Reinforcement Learning** model using **Proximal Policy Optimization** and **Soft Action Critic** algorithms
- Reduced lap time of virtual racing model by maximising speed vs constraints and placed **29th** globally against **2200 teams**

Technical Skills		
Programming Languages	Softwares and Libraries	Utilities and Tools
C, C++, Python, Java, JavaScript, MySQL, Bash, HTML, CSS, Dart	Flutter, React, Node.js, Flask, Firebase, Keras, Matplotlib, Pandas, sklearn	Git, GitHub, Linux, L ^A T _E X, AutoCAD, AutoDesk Inventor, Colab, Jupyter, Vim

Relevant Courses

(*ongoing)

Data Structures and Algorithms* Fundamentals of Computing Introduction to Logic	Introduction to Critical Thinking Computational Sociology* Introduction to Electronics	Introduction to Electrical Engineering Linear algebra and Differential Equations Theory of Mechanisms and Machines*
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Positions of Responsibility

Team Member | ViSiON

(Jun'22 - Feb'23)

- Collaborated closely with **11** dedicated team members on Path Planning, Reinforcement Learning, SLAM and Computer Vision
- Participated in global competitions like **AWS DeepRacer Student League, BARN Challenge, F1 Tenth** and **Santa 22**

Company Co-ordinator | Students' Placement Office

(Jun'22 - Mar'23)

- Collaborated in a **4-tier team** to conduct **Internship Drive** for **1100+** students and **Placement Drive** for **1400+** students
- Co-ordinated with companies like **Graviton, QuantBox** and **Quadeye** to hold placement and internship tests and interviews

Extra-Curricular Activities

- Organized research talks, **Integration Bee'22** and Stamatics' flagship event **Mathemania'22** as **Executive** in Stamatics
- Helped **6 freshmen** in adjusting to institute's environment on a social, academic and emotional front as a **Student Guide**
- Mentored **100+**Y22 Students in first year courses - '**Electrodynamics**' & '**Quantum Physics**' as an **Academic Mentor**
- Won **First Prize** in **Pitcher's Venture** conducted in honor of World Entrepreneurs Day by Entrepreneurship Cell, IIT Kanpur